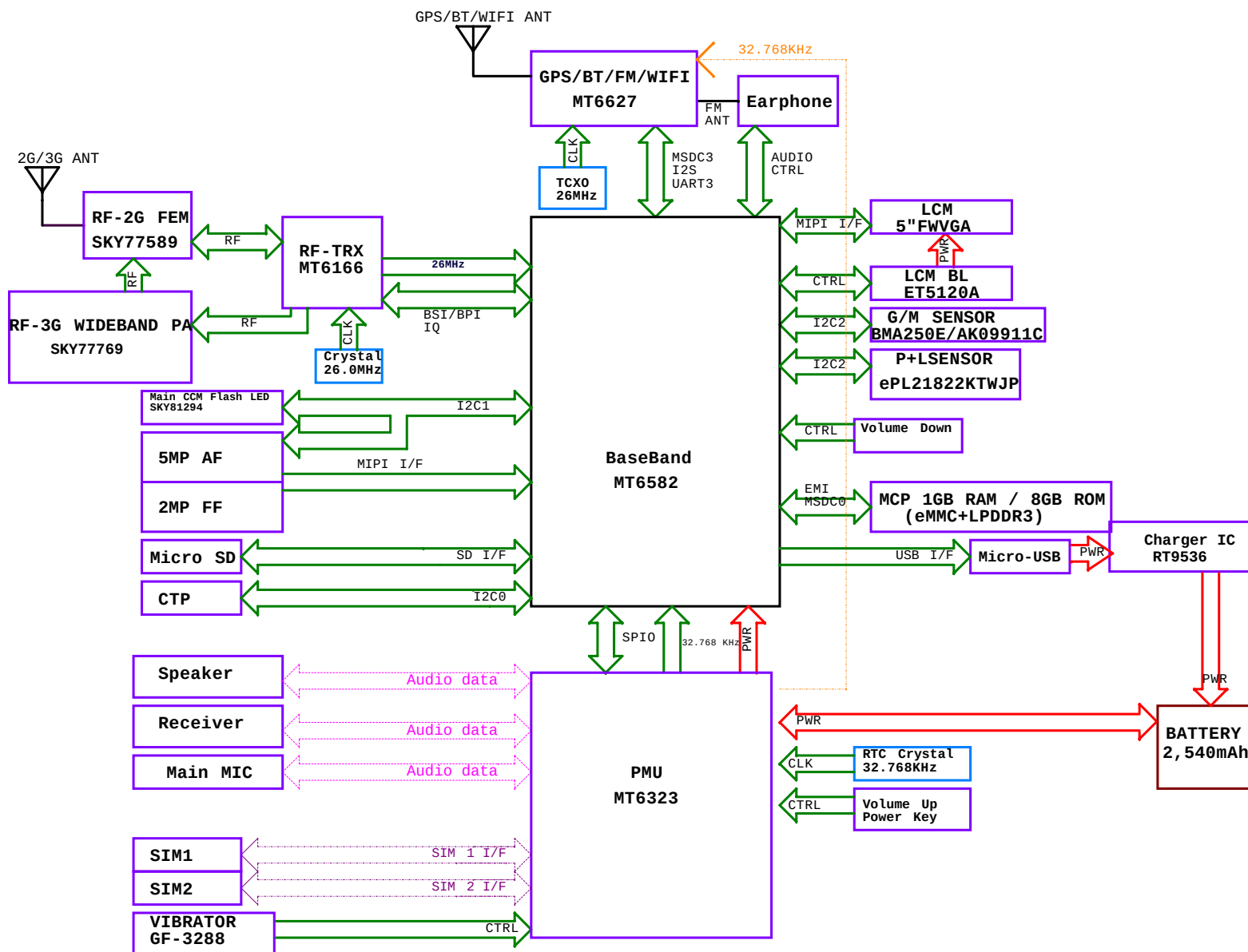




Change Notes

1. Added new features and improvements.

2. Fixed bugs and issues.

3. Updated dependencies and libraries.

4. Added new tests and documentation.

5. Improved performance and scalability.

6. Added new features and improvements.

7. Fixed bugs and issues.

8. Updated dependencies and libraries.

9. Added new tests and documentation.

10. Improved performance and scalability.

11. Added new features and improvements.

12. Fixed bugs and issues.

13. Updated dependencies and libraries.

14. Added new tests and documentation.

15. Improved performance and scalability.

16. Added new features and improvements.

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80. Improved performance and scalability.

81. Added new features and improvements.

82. Fixed bugs and issues.

83. Updated dependencies and libraries.

84. Added new tests and documentation.

85. Improved performance and scalability.

86. Added new features and improvements.

87. Fixed bugs and issues.

88. Updated dependencies and libraries.

89. Added new tests and documentation.

90. Improved performance and scalability.

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94. Added new tests and documentation.

95. Improved performance and scalability.

96. Added new features and improvements.

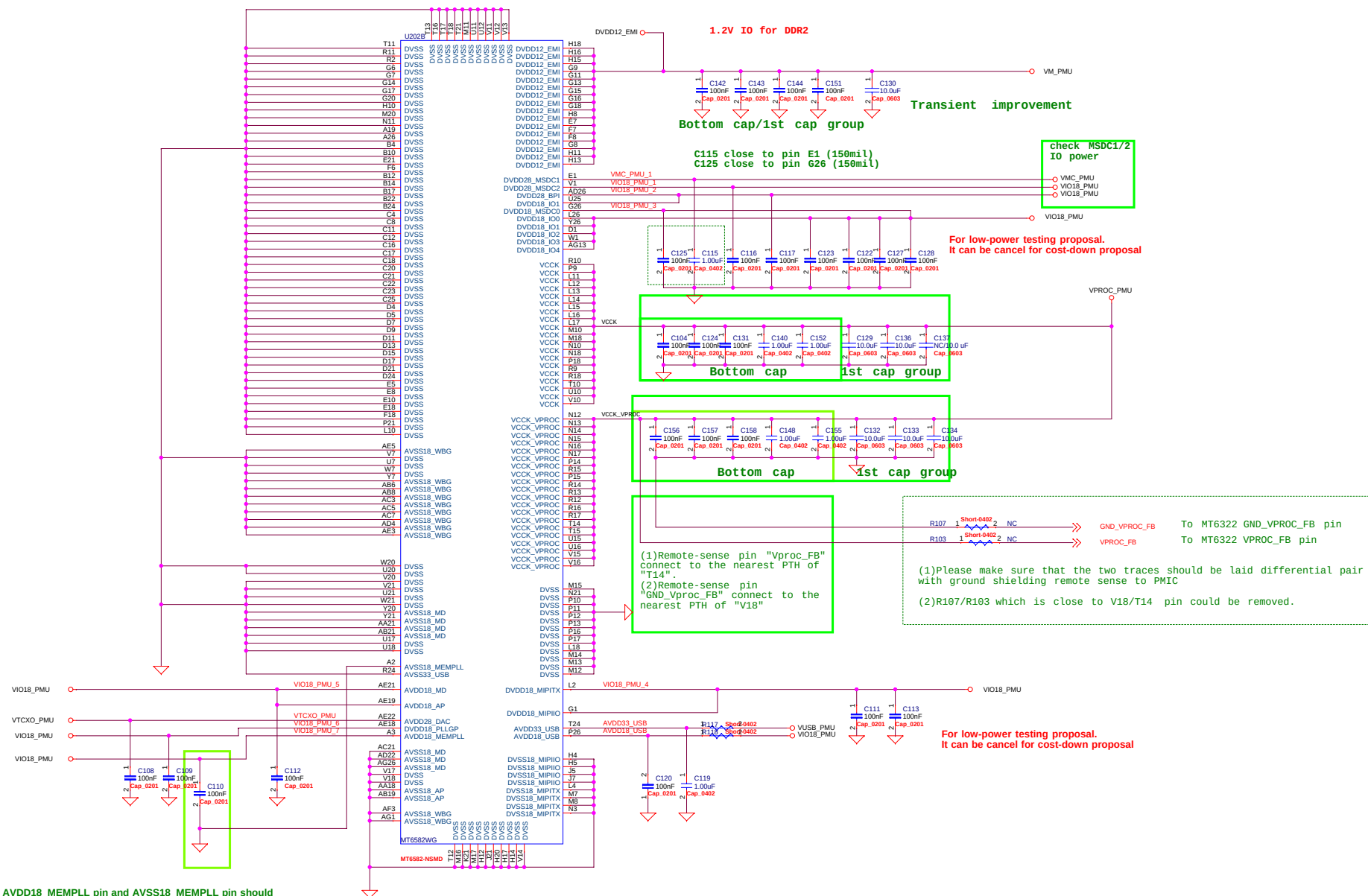
97. Fixed bugs and issues.

98. Updated dependencies and libraries.

99. Added new tests and documentation.

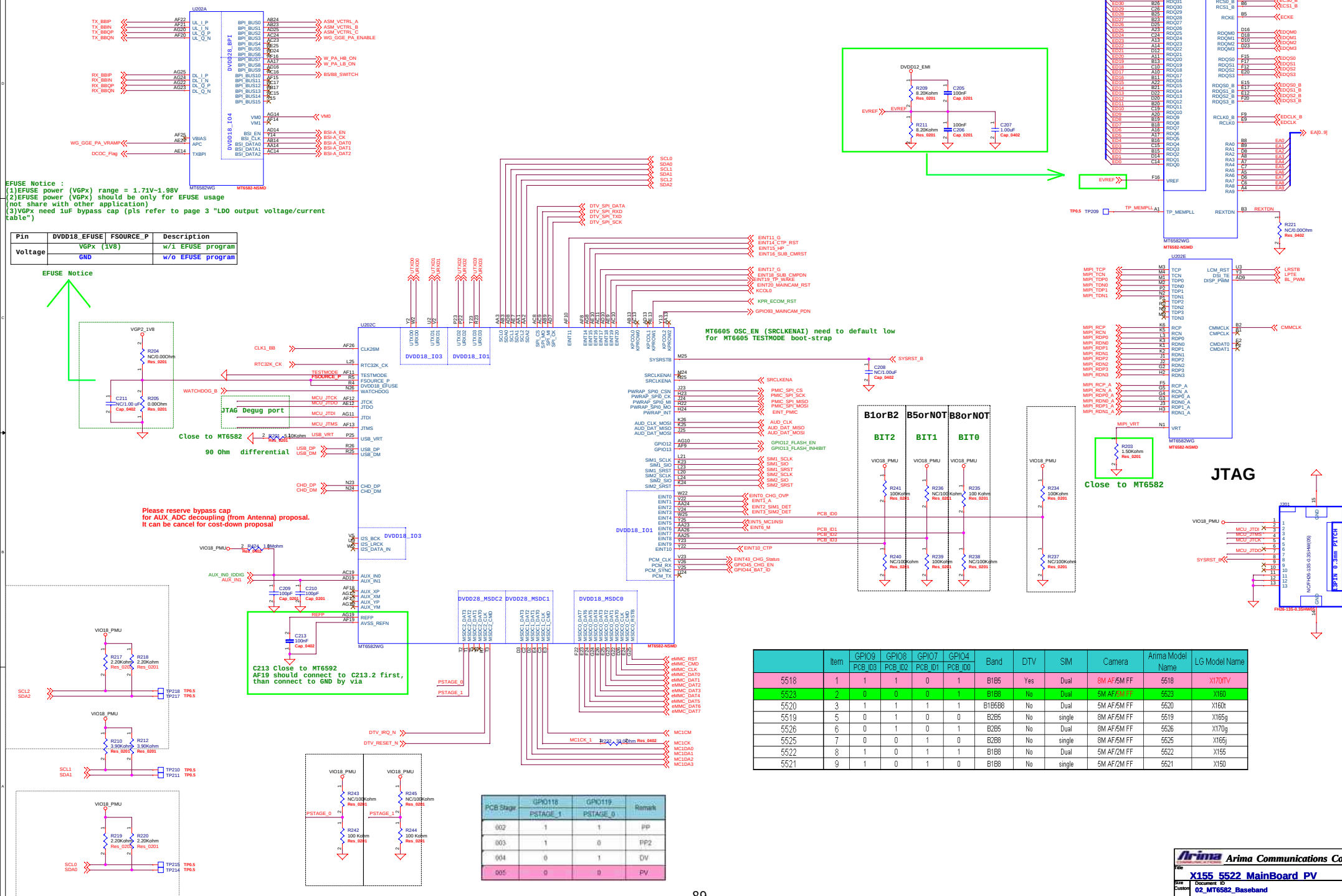
100. Improved performance and scalability.

MT6582_Power



AVDD18_MEMPLL pin and AVSS18_MEMPLL pin should be connected with PCB CAP first, and then connected with PMU and PCB ground. CAP should be near MT6582 as possible as we can.

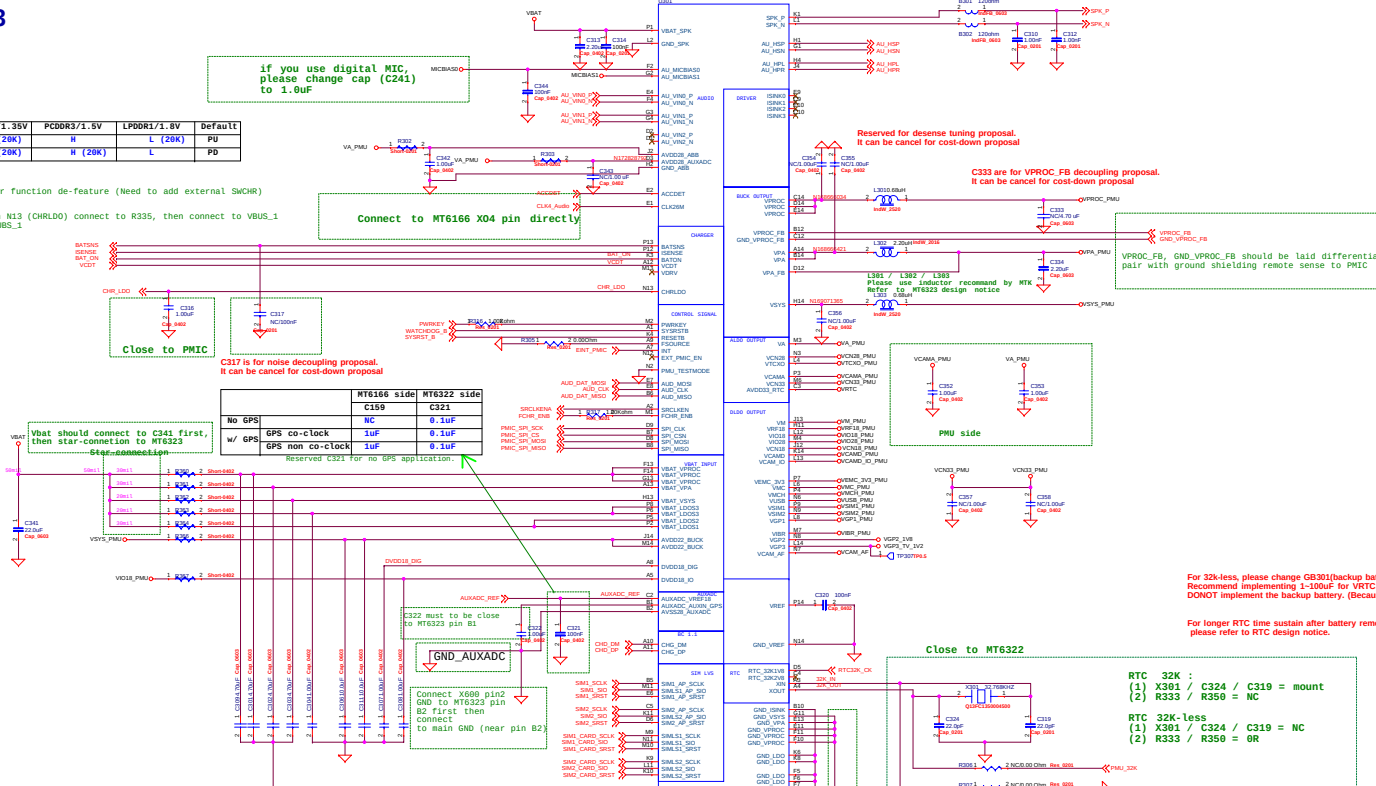
MT6582_Baseband



PMIC_MT6323

Symbol	LP06R2/1.2V	PC06R3/1.35V	PC06R3/1.5V	LP06R1/1.8V	Default
SPI_CS#	N	L (20K)	N	L (20K)	PU
AUX_R0S1	L	N (20K)	N (20K)	L	PU

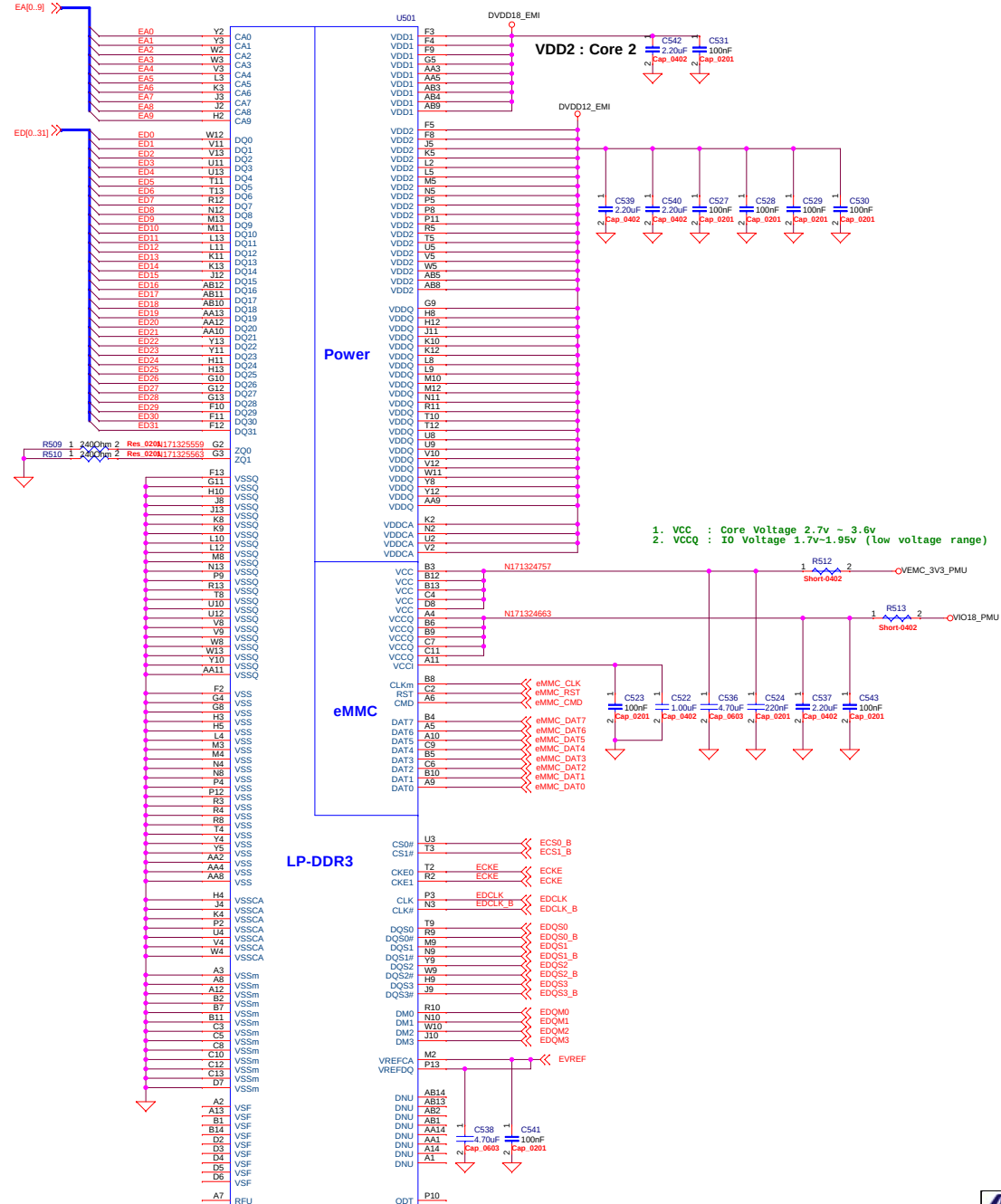
MT6323 switching charger function de-feature (Need to add external SWCHR)
Change notice:
Page 3
1. Add R335, MT6323 pin R13 (CHRLDO) connect to R335, then connect to VBUS_1
2. R330.1 connect to VBUS_1



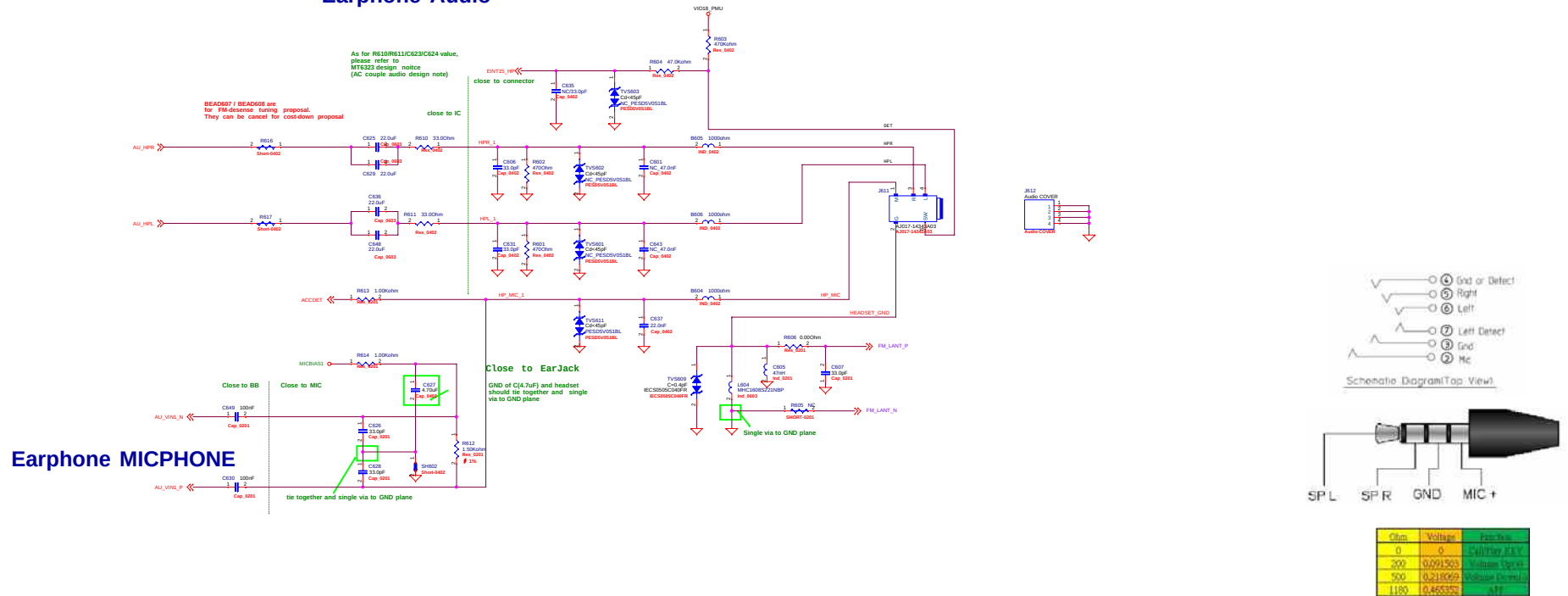
Regulator	Output Voltage(V)	Output Current(mA)	Input Decoupling	Output Decoupling	Notes
VPROC	0.7-1.4	2800	>10uF	L=0.68uH, C=10uF*4	Total output cap>40uF
VSYS	2.2	1200	>10uF	L=0.68uH, C=10uF*2	Total output cap>20uF
VPA	0.5-3.4	600	>4.7uF	L=2.2uH, C=2.2uF+2.2uF	Output cap range 4.4uF +/-20%
LDO	Output Voltage(V)	Output Current(mA)	Input Decoupling	Output Decoupling	Notes
VM	1.24/1.39/1.54/1.84	700	10uF	-20%~+20%	Far-end bypass cap
VRF18	1.825	200	1uF	-20%~+200%	Far-end bypass cap
VT018	1.8	300	4.7uF	-20%~+200%	Far-end bypass cap
VCN18	1.8	120	1uF	-20%~+20%	Far-end bypass cap
VCAMD	1.2 / 1.3/1.5/1.8	150	1uF	-20%~+20%	Far-end bypass cap
VCAM_IO	1.8	100	1uF	-20%~+20%	Far-end bypass cap
VGP3	1.2 / 1.3/1.5/1.8	200	1uF	-20%~+20%	Far-end bypass cap
VA	2.8	150	1uF	-20%~+20%	Far-end bypass cap
VTCKO	2.8	40	1uF	-20%~+20%	Far-end bypass cap
VCN28	2.8	30	1uF	-20%~+20%	Far-end bypass cap
VCAMA	2.8	150	3.2uF	-20%~+20%	1uF near-end, 2.2uF Far-end bypass cap
VCN33	3.3 / 3.4/3.5/3.6	240(MT6323) 350(MT6322)	4.7uF	-20%~+20%	Far-end bypass cap
VT028	2.8	200	2.2uF	-20%~+200%	Far-end bypass cap
VUSB	3.3	20	1uF	-20%~+20%	Far-end bypass cap
VMC	1.8 / 3.3	100	1uF	-20%~+20%	Far-end bypass cap
VMCH	3.0 / 3.3	400	2.2uF	-20%~+20%	Far-end bypass cap
VEMC_3V3	3.0 / 3.3	400	4.7uF	-20%~+20%	Far-end bypass cap
VCAM_AF	1.2/1.3/1.5/1.8/2.0 2.8/3.0/3.3	100	1uF	-20%~+20%	Far-end bypass cap
VSIM1	1.8 / 3.0	50	1uF	-20%~+20%	Far-end bypass cap
VSIM2	1.8 / 3.0	50	1uF	-20%~+20%	Far-end bypass cap
VGP1	1.2/1.3/1.5/1.8/2.0 2.8/3.0/3.3	100	1uF	-20%~+20%	Far-end bypass cap
VGP2	1.2/1.3/1.5/1.8/2.0 2.5/2.8/3.0	100	1uF	-20%~+20%	Far-end bypass cap
VTBR	1.2/1.3/1.5/1.8/2.0 2.8/3.0/3.3	100	1uF	-20%~+20%	Far-end bypass cap
VDIG18	1.8	20	1uF	-20%~+20%	Far-end bypass cap
VRTC	2.8	2	0.1uF to 1000uF	-20%~+20%	Far-end bypass cap

eMMC+LPDDR3

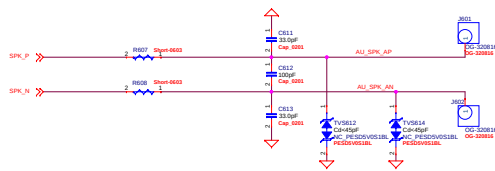
221 Ball, 0.5mm pitch
LPDDR3 eMMC
VDD1=1.8V VCC=3.3V
VDD2=1.2V VCCQ=1.8V
VDDCA=1.2V
VDDQ= 1.2V



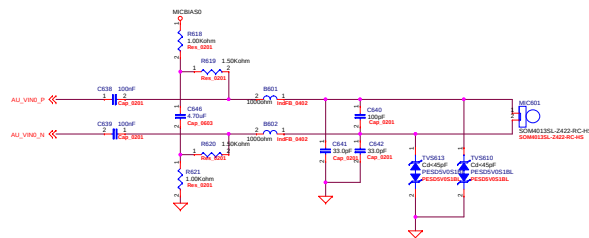
Earphone Audio



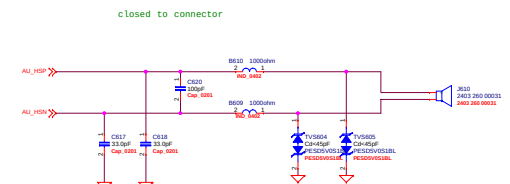
SPEAKER



Main Microphone



RECEIVER



D

C

B

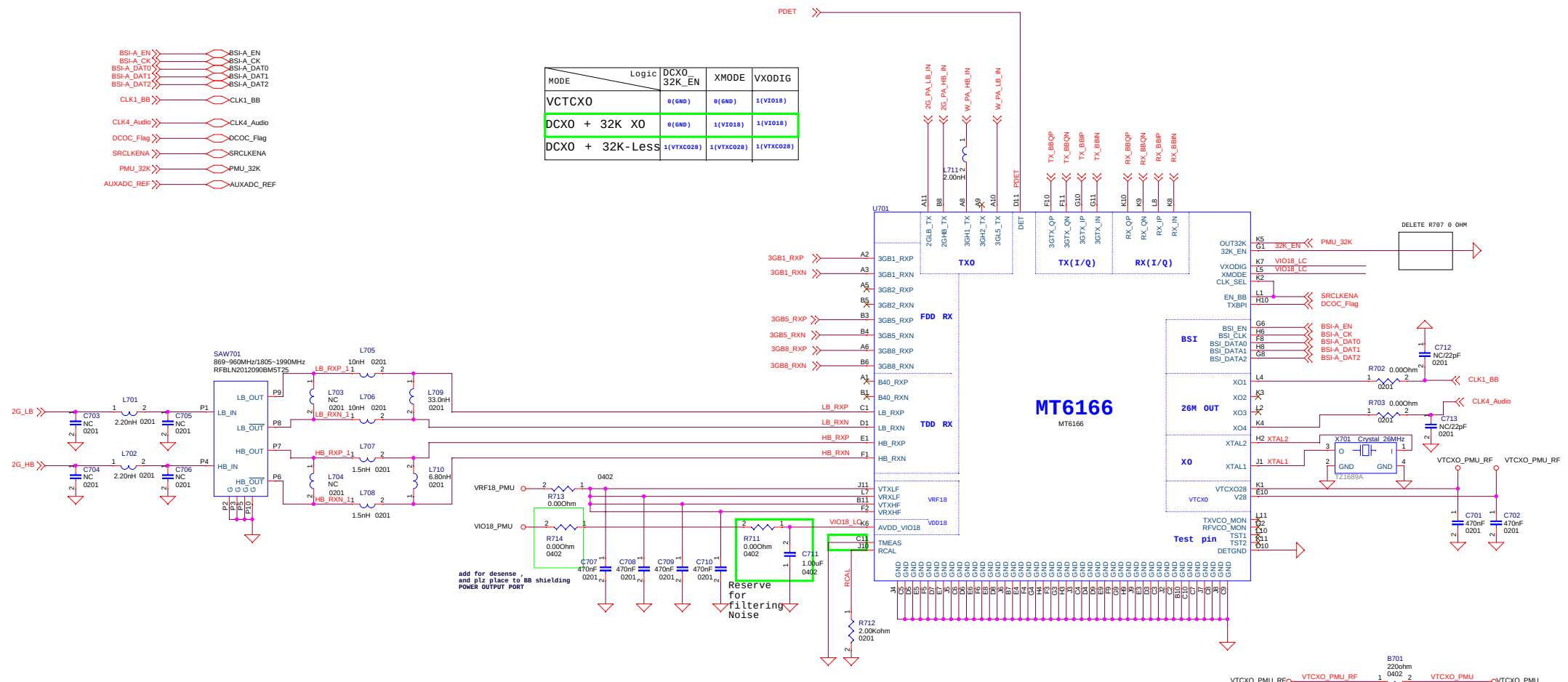
A

D

C

B

A



Remove external thermistor;
use MT6166 built-in
thermistor

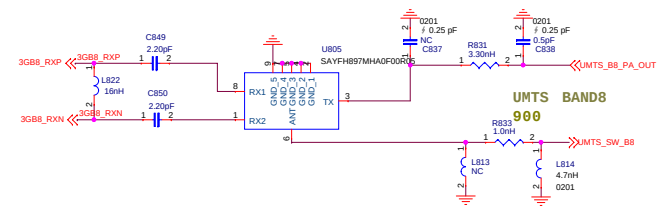
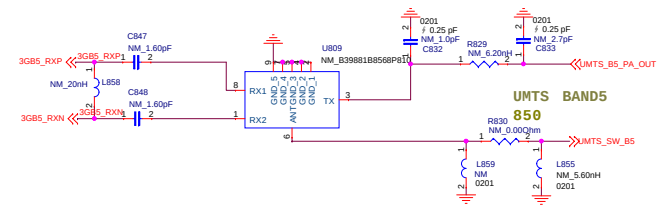
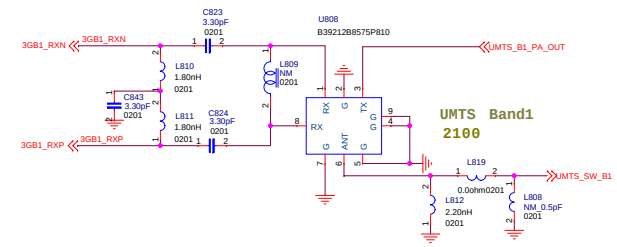
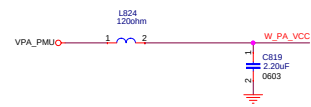
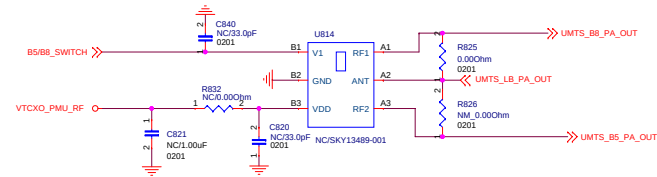
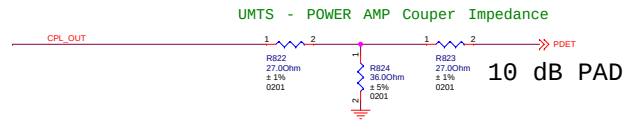
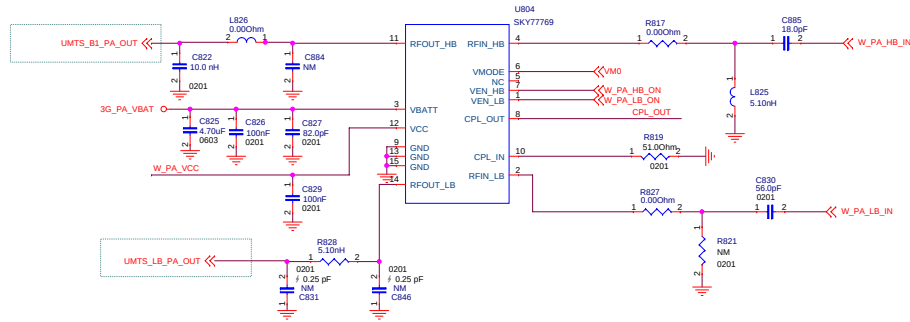
Net name	Pin name	Power domain	Max Current consumption (mA)
RXHF	F2	VRF18	100
VTCX028	K1	VTCX0	6
VRXLF	L7	VRF18	35
VTXLF	J11	VRF18	35
V28	E10	VTCX0	1
VTXHF	B11	VRF18	125
VIO18	K6	VIO18	1

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Size A3 Document ID
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UMTS -WINDBAND POWER AMP



FEM

Ant. Spring

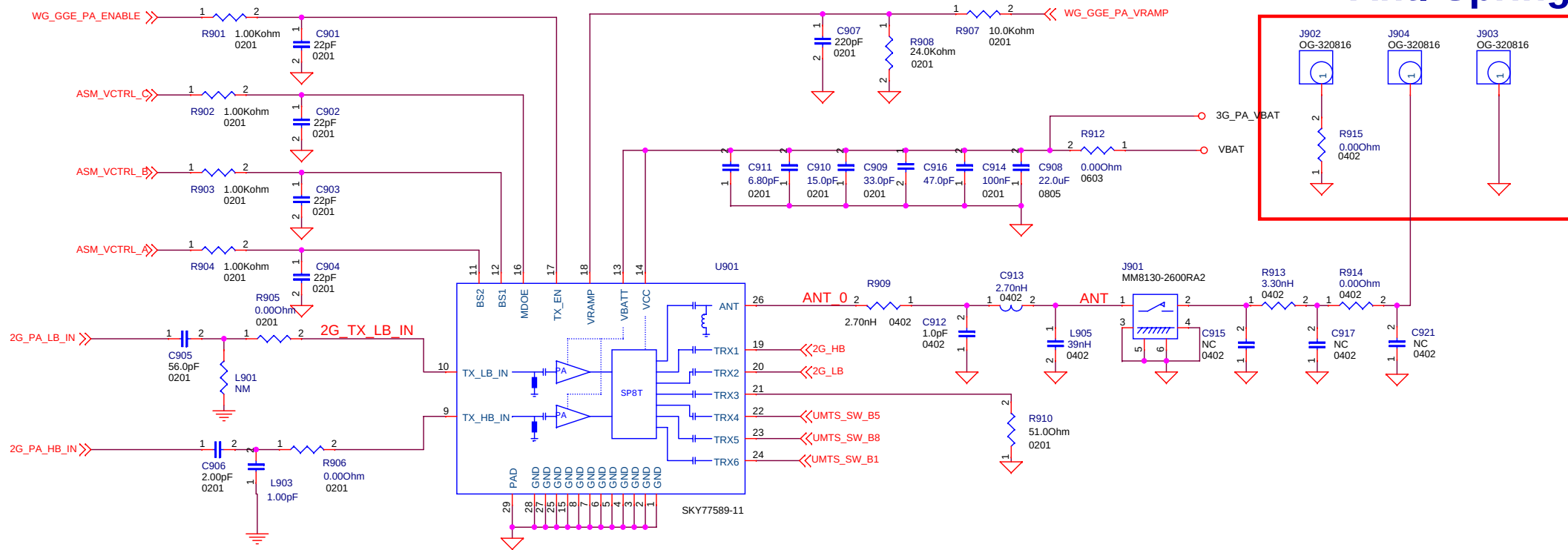


Table 3. SKY77589-11 Mode Control Logic

Mode	Input Control Bits			
	TdN	MODE	BS1	BS2
Standby	0	0	0	0
LB_GMSK_Tx	1	0	0	1
HB_GMSK_Tx	1	0	1	1
Trx1	0	1	0	0
Trx2	0	1	1	0
Trx3	0	1	0	1
Trx4	0	1	1	1
Trx5	0	0	1	0
Trx6	0	0	0	1

GPS xLNA

Close to ANT

matching value depends on LNA selected

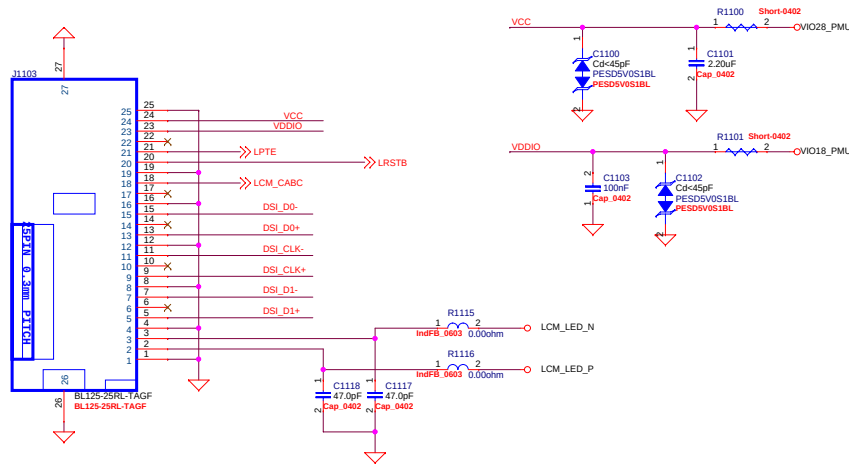
Bead1001 are for FM desense-free proposal.
They can be cancel for cost-down proposal

Close to MT6627

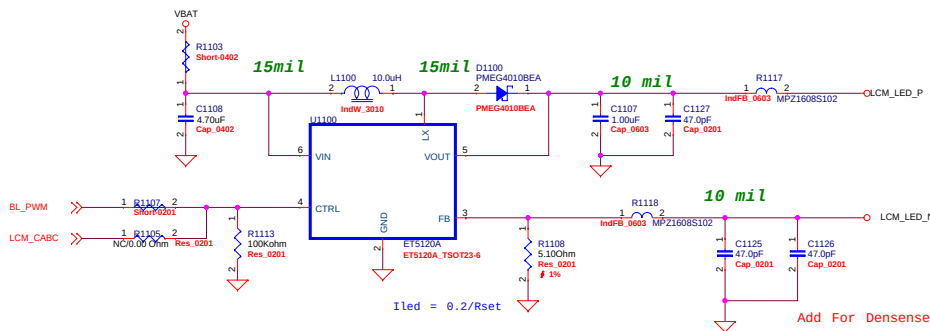
For MMD TP

MODE	Component	X1001	R1015	R1016	R1014
Xtal Mode		QVL TCXO	0ohm	NC	
GPS co-clock		NC	NC	0 ohm	

LCM



Backlight LED DRV



Capacitive Touch Panel

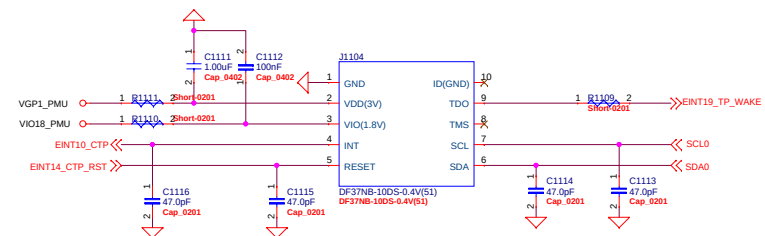
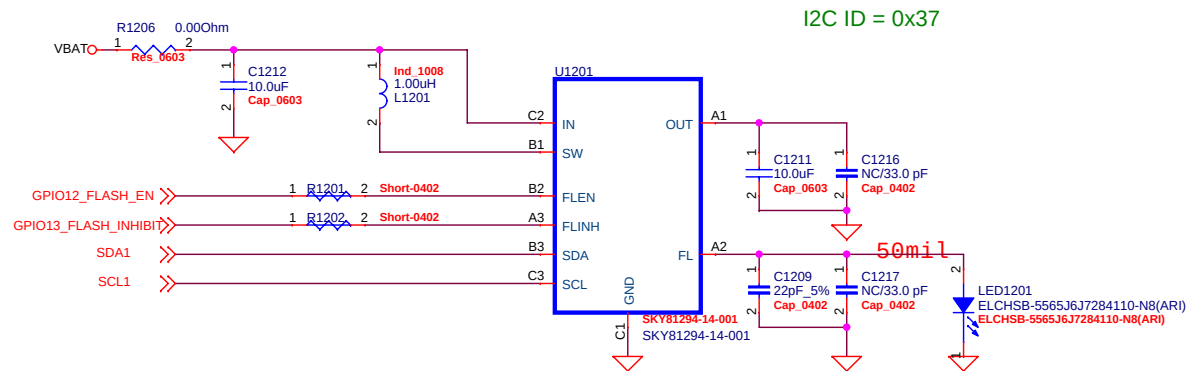


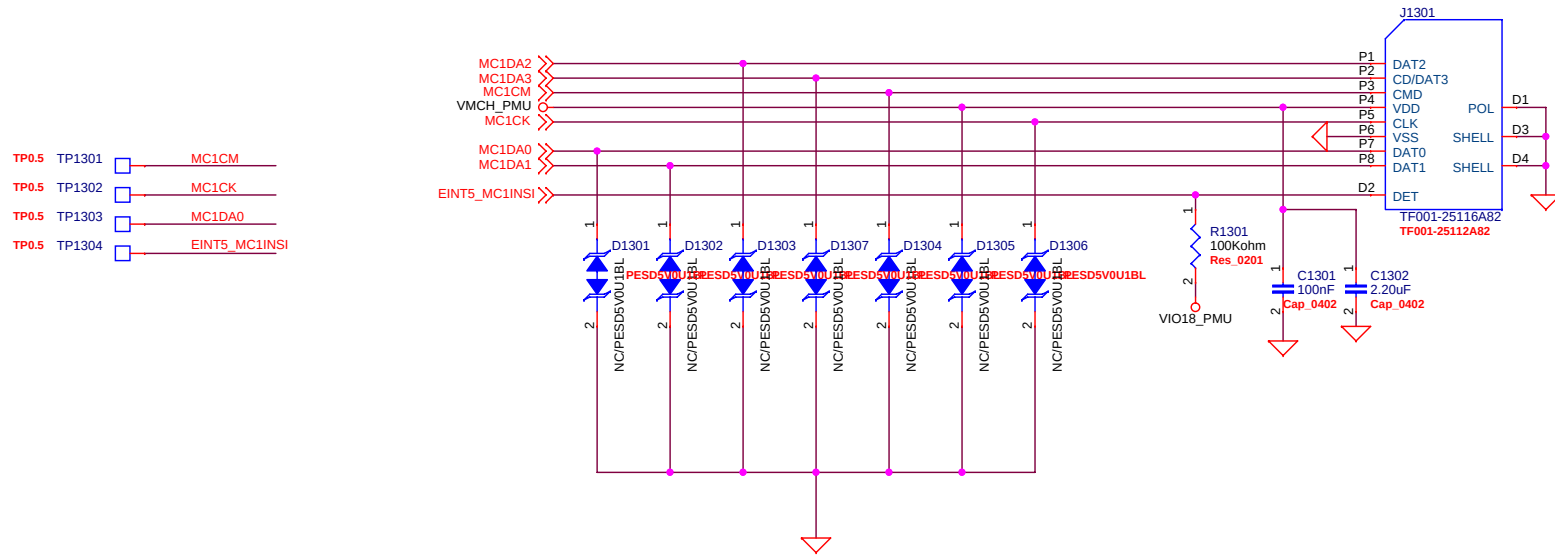
Figure 10 shows the pin connections for the ACFT4A2G900E component, specifically for the MIPI signals. The component is shown in five different configurations, each with a 100MHz clock signal. The pins are labeled FL1201, FL1202, FL1203, FL1204, and FL1205. The connections are as follows:

- FL1201 100MHz:** MIPI_RCP (pin 2), MIPI_RCN (pin 1), DCKP (pin 3), DCKN (pin 4).
- FL1202 100MHz:** MIPI_RDP0 (pin 1), MIPI_RDN0 (pin 2), DMO0P (pin 4), DMO0N (pin 3).
- FL1203 100MHz:** MIPI_RDP1 (pin 2), MIPI_RDN1 (pin 1), DMO1P (pin 3), DMO1N (pin 4).
- FL1204 100MHz:** MIPI_RDP2 (pin 2), MIPI_RDN2 (pin 1), DMO2P (pin 3), DMO2N (pin 4).
- FL1205 100MHz:** MIPI_RDP3 (pin 1), MIPI_RDN3 (pin 2), DMO3P (pin 4), DMO3N (pin 3).

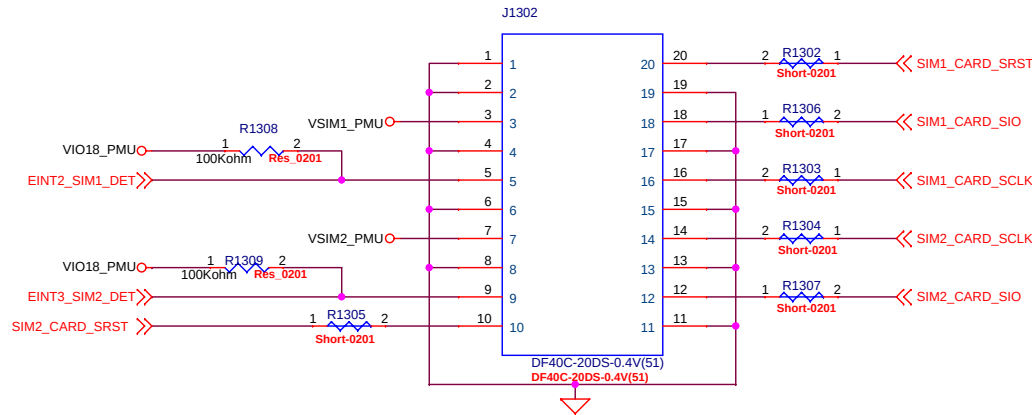


The schematic diagram illustrates the connection of the VIB1201 component to the VIB PMU. The VIB1201 component is connected to the VIB PMU via a 15mil trace. The VIB1201 component is connected to the VIB PMU via a 15mil trace. The VIB1201 component is connected to the VIB PMU via a 15mil trace.

mSD CARD



SIM Board I/F



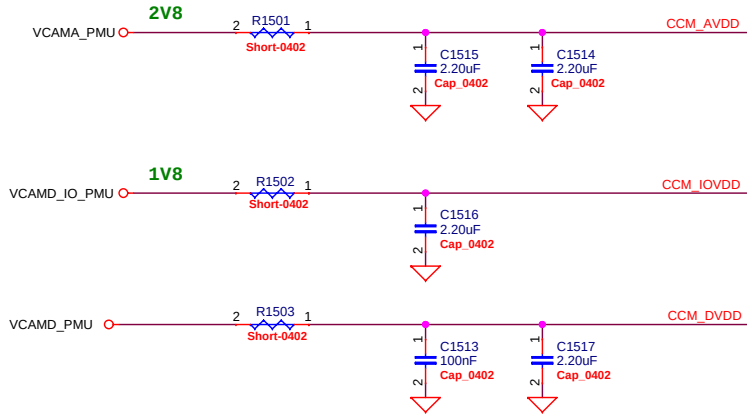
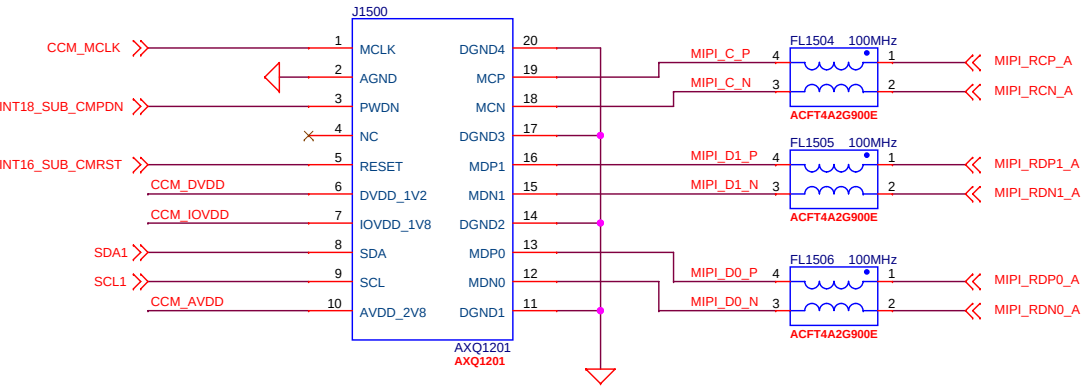
		Arima Communications Corp.	
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X155 5522 MainBoard PV			
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B	13_Memory_CARD_DSIM_BT0B	PA1	
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Close to MT6323 AUXADC_VREF18 PIN,
and the path need shielding with GND

[illegible][illegible]

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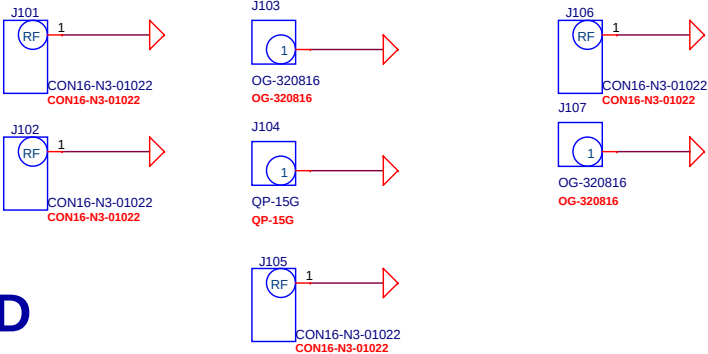
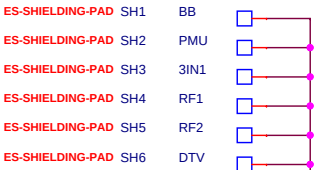
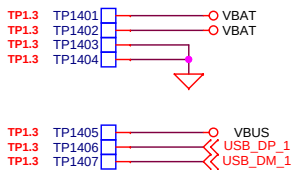
SUB Camera Module 2M/5M Compatible



Test point

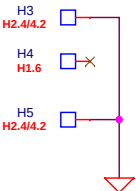
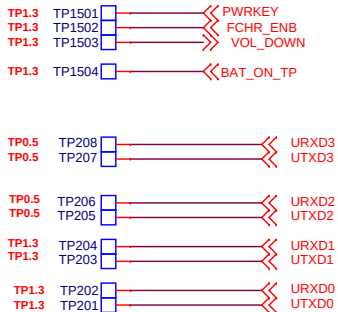
Shelding case

Spring Connector



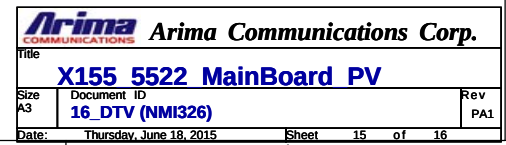
Screw Hole

FID

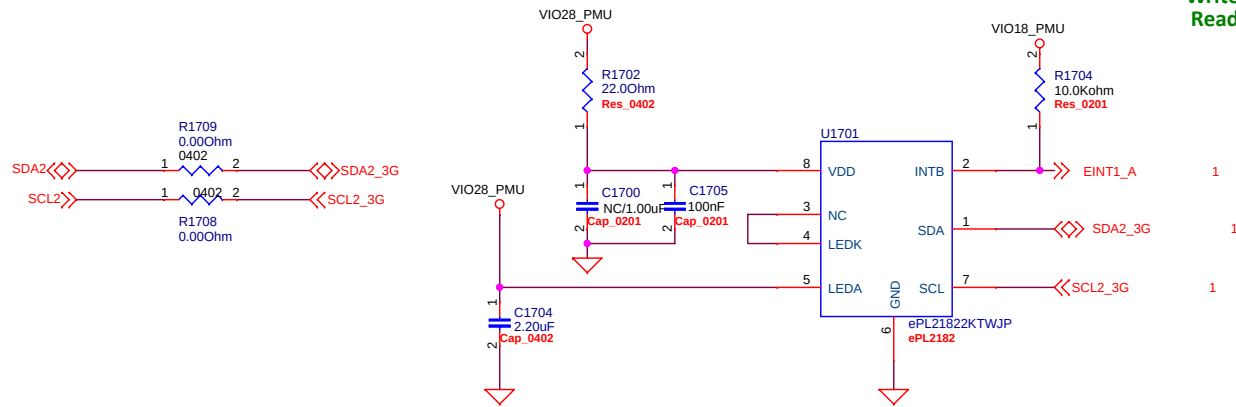


Arima Communications Corp.			
Title			
X155 5522 MainBoard PV			
Size	Document ID	Rev	
B	15_Sub Camera_Test point	PA1	
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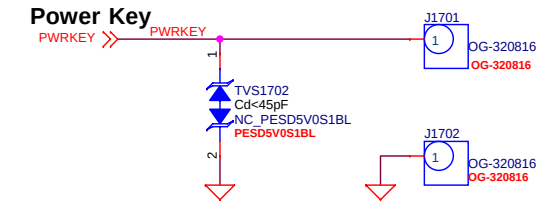
ANT



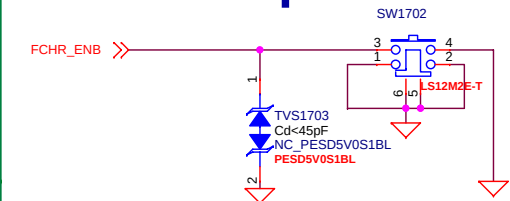
3 in 1 ALS+PS+IR



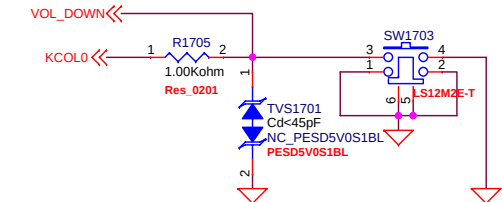
Power Key



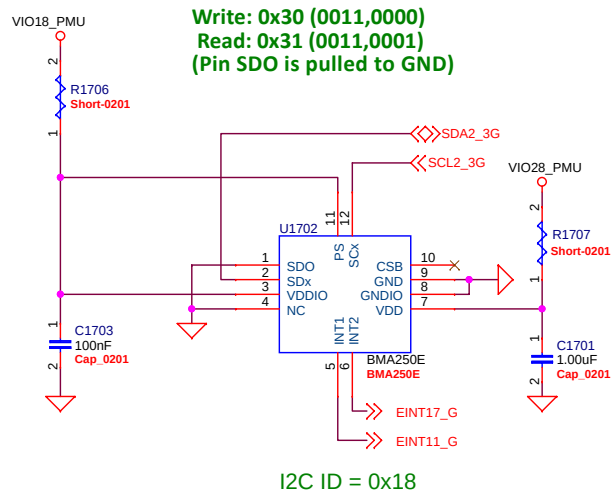
Volume Up



Volume Down

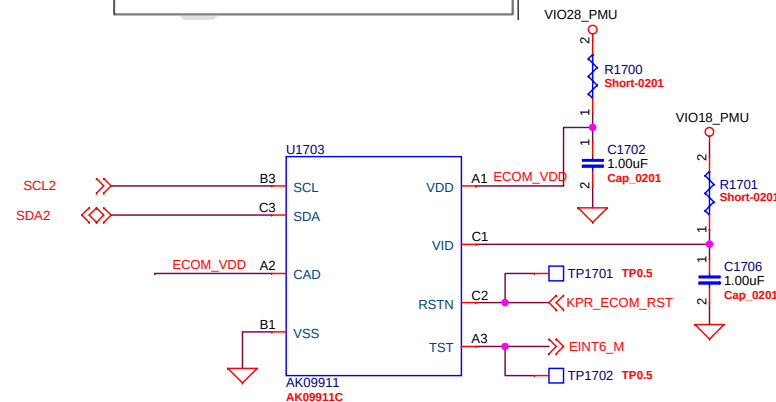


G-Sensor



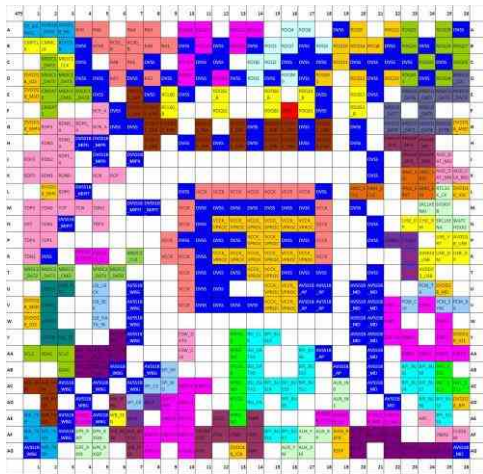
eCompass

Slave address select	
CAD	address
VSS	0 0 0 1 1 0 0 R/W
VDD	0 0 0 1 1 0 1 R/W



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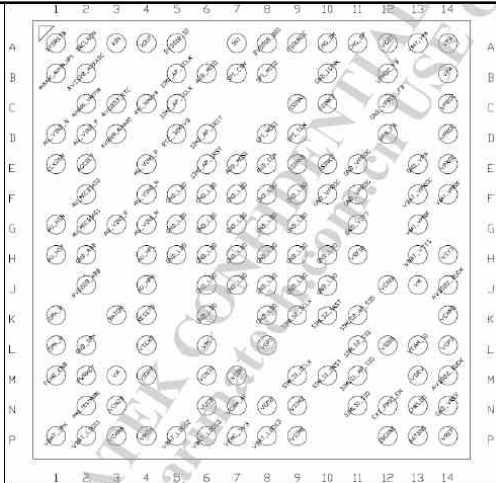
U202_MT6582_BB chipIC,Digital Baseband Processor(Top)



U501_KMQ7X000SA-B315,IC,MCP. eMMC+LPDDR3(Top)



U301_MT6323_IC,PMIC (Top)



U701_MT6166V_IC ,Transceiver (Top)

